

PROJECT ROADMAP AND ANALYTICAL FRAMEWORK

DATA ANALYTICS AND VISUALIZATION - 24DSB3101

Intelligent Customer Churn Prediction Using Data Analytics and Classification Models

S. No.	University ID	Name
1	2420030540	Allipuram Akhil
2	2420030440	Devineni Surya

Guide: DR. K. Swapnika

Project Focus

A data analytics and machine learning framework for predicting customer churn, understanding churn factors, assigning churn-risk levels, and presenting interpretable results through an interactive dashboard.

Core Pipeline

- Data collection and quality assessment
- Exploratory Data Analysis (EDA)
- Preprocessing and class balancing
- Feature engineering and selection
- Classification model training and comparison
- Churn probability and risk levels
- SHAP explainability
- Interactive dashboard

PROJECT ABSTRACT AND OBJECTIVES

Abstract

Customer churn is a major challenge for organizations, as losing existing customers can negatively affect customer relationships and business performance. Machine learning models can predict whether a customer is likely to churn by analyzing demographic, service, usage, contract, and billing-related information. This project proposes a data analytics and machine learning framework for intelligent customer churn prediction by integrating EDA, preprocessing, feature engineering, classification models, risk scoring, explainability, and visualization.

The customer churn dataset undergoes missing-value handling, categorical encoding, normalization, and class balancing using SMOTE. Detailed EDA is performed to understand churn distribution, customer patterns, correlations, outliers, and important relationships. Feature engineering and feature selection identify relevant customer, service, usage, contract, and billing factors.

Logistic Regression, Decision Tree, Random Forest, XGBoost, and CatBoost are trained and evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Models generate churn probabilities that are categorized into Low, Medium, and High churn-risk levels.

SHAP is used to identify influential factors contributing to churn, providing global feature importance and local explanations for individual customer predictions. Results are presented through visualizations and an interactive dashboard containing model performance, churn factors, risk categories, and prediction results.

Objectives

- Perform systematic EDA on customer churn data.
- Apply preprocessing, encoding, normalization, and SMOTE.
- Train and compare five classification models.
- Evaluate using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
- Generate churn probabilities and Low/Medium/High risk levels.
- Use SHAP for global and individual prediction explanations.
- Present results through an interactive dashboard.

DATA ANALYTICS APPROACH - EDA PIPELINE

EDA Flow

Raw Customer Churn Dataset

- ↓ Dataset Overview
- ↓ Data Quality Assessment
 - ↓ Descriptive Statistics
 - ↓ Missing Value Analysis
 - ↓ Duplicate Detection
 - ↓ Outlier Detection
- ↓ Univariate Analysis
 - ↓ Bivariate Analysis
- ↓ Multivariate Analysis
 - ↓ Correlation Analysis
- ↓ Feature Engineering
 - ↓ Feature Selection
- ↓ Machine Learning Models

Customer Churn EDA Focus

- Understand churned versus retained customers.
- Analyze demographic, service, usage, contract, and billing characteristics.
- Identify missing values, duplicates, inconsistencies, and outliers.
- Study relationships between customer attributes and churn.
- Analyze correlations and possible multicollinearity.
- Create meaningful derived features where supported by the dataset.
- Select relevant features for classification and explainability.

EDA TECHNIQUES TO IMPLEMENT

S.No.	Technique	Purpose	Visualization / Output
1	Dataset Overview	Understand structure	info(), describe()
2	Data Quality Assessment	Check data integrity	Quality report
3	Descriptive Statistics	Summarize features	Summary tables
4	Missing Value Analysis	Identify missing data	Bar plots / heatmap
5	Duplicate Records	Detect redundancy	Duplicate report
6	Outlier Detection	Identify anomalies	Boxplots / IQR
7	Univariate Analysis	Study distributions	Histograms / KDE
8	Bivariate Analysis	Study relationships	Bar / scatter plots
9	Multivariate Analysis	Study combined patterns	Pair plots
10	Correlation Analysis	Identify relationships	Correlation heatmap
11	Feature Engineering	Create useful features	Feature distributions
12	Feature Selection	Retain relevant predictors	Importance ranking

Key Churn Questions

- Which customer characteristics are associated with higher churn?
- How do contract and service characteristics relate to churn?
- How do billing and usage variables differ between churned and retained customers?
- Which features contribute most to the final models?

PREPROCESSING, CLASSIFICATION AND EXPLAINABILITY

Preprocessing

- Missing-value handling
- Categorical data encoding
- Numerical feature normalization / scaling where required
- Class imbalance handling using SMOTE
- Train-test split

Machine Learning Models

- Logistic Regression - interpretable baseline
- Decision Tree - rule-based classification
- Random Forest - ensemble classification
- XGBoost - gradient boosting classification
- CatBoost - boosting model suited to categorical features

Explainability using SHAP

- Global explanation: most influential churn-related features.
- Local explanation: why an individual customer receives a churn prediction.
- Positive SHAP contribution increases predicted churn risk.
- Negative SHAP contribution decreases predicted churn risk.
- Feature contributions can be visualized for stakeholders.

Risk Scoring

Output	Meaning
Churn Probability	Model-estimated probability of churn
Low Risk	Lower predicted likelihood of churn
Medium Risk	Moderate predicted likelihood of churn
High Risk	Higher predicted likelihood of churn

VISUALIZATION AND SYSTEM ARCHITECTURE

Planned Visualizations

- Churn distribution
- Data-quality and missing-value plots
- Boxplots and outlier analysis
- Correlation heatmap
- Feature importance
- ROC curves
- Confusion matrices
- Model comparison charts
- SHAP global importance
- SHAP local explanations
- Interactive churn-risk dashboard

Proposed Architecture

Layer	Components
Data Layer	Customer Churn Dataset
Preprocessing Layer	Missing Values → Encoding → Scaling → SMOTE → Train/Test Split
Analytics Layer	EDA → Feature Engineering → Feature Selection
Modeling Layer	Logistic Regression Decision Tree Random Forest XGBoost CatBoost
Prediction Layer	Churn Probability → Low / Medium / High Risk
Explainability Layer	SHAP Global Importance → SHAP Local Explanation
Output Layer	Interactive Dashboard → Model Results → Risk Categories → Churn Factors

COMPARATIVE ANALYSIS AND RESEARCH PIPELINE

Models for Comparative Analysis

Model	Evaluation Metrics
Logistic Regression	Accuracy, Precision, Recall, F1-Score, ROC-AUC
Decision Tree	Accuracy, Precision, Recall, F1-Score, ROC-AUC
Random Forest	Accuracy, Precision, Recall, F1-Score, ROC-AUC
XGBoost	Accuracy, Precision, Recall, F1-Score, ROC-AUC
CatBoost	Accuracy, Precision, Recall, F1-Score, ROC-AUC

Research Pipeline

Phase 1: Data Collection and Quality Assessment

↓ Cleaning and Preprocessing

Phase 2: Exploratory Data Analysis

↓ 12+ EDA Techniques

↓ Feature Engineering

↓ Feature Selection

Phase 3: Classification Modeling

↓ Train-Test Split

↓ Five Classification Models

Phase 4: Evaluation and Explainability

↓ Accuracy, Precision, Recall, F1-Score, ROC-AUC

↓ SHAP Global and Local Explanations

Phase 5: Output

↓ Churn Probability and Risk Categories

↓ Interactive Dashboard

↓ Comparative Analysis and Conclusions

IMPLEMENTATION ROADMAP AND TECHNOLOGY STACK

Implementation Roadmap

Week	Activities	Deliverables
Week 1	Data Collection & Preprocessing	Clean dataset, preprocessing pipeline, initial EDA
Week 2	Complete EDA Pipeline	EDA report and visualizations
Week 3	ML Model Training	Five trained models and initial evaluation
Week 4	SHAP Explainability	Global and local explanation visualizations
Week 5	Dashboard & Comparative Analysis	Working dashboard, model comparison, final report

Technology Stack

Category	Tools / Libraries
Data Processing	Pandas, NumPy, Scikit-learn
EDA & Visualization	Matplotlib, Seaborn, Plotly
Class Imbalance	imbalanced-learn / SMOTE
Machine Learning	Scikit-learn, XGBoost, CatBoost
Explainable AI	SHAP
Dashboard	Streamlit
Development	Jupyter Notebook / Google Colab / Python

PROJECT SUMMARY AND TEAM INFORMATION

Expected Project Output

- Cleaned and analyzed customer churn dataset.
- Complete EDA pipeline with documented visualizations.
- Five trained classification models.
- Comparative evaluation using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
- Churn probabilities and Low, Medium, High risk categories.
- SHAP-based global and local explanations.
- Interactive dashboard for prediction and analysis.
- Final comparative analysis for data-driven customer retention decisions.

Team Information

S. No.	University ID	Name
1	2420030540	Allipuram Akhil
2	2420030440	Devineni Surya

Project Title

Intelligent Customer Churn Prediction Using Data Analytics and Classification Models

Guide	HOD
DR. K. Swapnika	_____